

check if large number is divisible by 7

using iterative Division - $O(\log_{10} n)$ Time and $O(1)$ Space

A number of the form $10a + b$ is divisible by 7 if and only if $a - 2b$ is divisible by 7. In other words, subtract twice the last digit from the number formed by the remaining digits

Continue until a small number

- Start by checking if the number is 0 or 7, as these are trivially divisible by 7.
- Iteratively remove the last digit and apply the rule: subtract twice the last digit from the remaining number.
- Repeat the process until a single-digit number is left, and check if it's 0 or 7 to confirm divisibility.

```
public static void main(String[] args)
{
    int n = 371;
    if (isDivBy7(Math.abs(n))) {
        System.out.println("True");
    }
    else {
        System.out.println("False");
    }
}
```

```
public static boolean isDivBy7(int n)
{
    if (n == 0 || n == 7)
        return true;

    while (n >= 10) {
        int lastD = n % 10;
        n = n / 10;
        n = Math.abs(n - 2 * lastD);
    }
    return n == 0 || n == 7;
}
```

How does this work?

Let 'b' be the last digit of a number 'n' and let 'a' be the number we get when we split off 'b'

The representation of the number may also be multiplied by any number relatively prime to the divisor without changing its divisibility

After observing that 7 divides 21, we can perform the following.

$$\begin{array}{lcl} 10a + b & & \\ \text{multiply by 2} & & \\ 20a + 2b & \Rightarrow & \text{Which can be written as} \\ & & 21a - a + 2b \\ & & \downarrow \\ \text{multiply by -1 gives} & \Leftarrow & \text{eliminating the multiple of 21} \\ a - 2b & & \text{gives} \\ & & -a + 2b \\ & & \boxed{a - 2b} \end{array}$$

Time Complexity: $O(\log_{10} n)$ The number of digits in the number is the no of times the iteration occurs.

If suppose the number given is very large like 10^{100000} then you can take input in the form of string and can use **a-2b** approach using addition and subtraction of strings, but then the complexity will be $O(n^2)$, where n is the length of the number.

Space Complexity: $O(1)$

Checking if a large Number is divisible by 9 or not

Input: $s = "69354"$

Output: Yes

Explanation: 69354 is divisible by 9.

Input: $s = "234567876799333"$

Output: No

Explanation: 234567876799333 is not divisible by 9.

Input: $s = "3635883959606670431112222"$

Output: No

Explanation: 3635883959606670431112222 is not divisible by 9.

How does this work?

Consider the number 1332. We can break it down as follows

$$69354 = 6 \times 10000 + 9 \times 1000 + 3 \times 100 + 5 \times 10 + 4$$

Key Observation

When dividing powers of 10 by 9, the remainder is always 1.
This means that Remainder of 10^i (for any i) when divided by 9 is 1.

This property allows us to simplify the number for divisibility by 9

Proof

Now, let's find the remainder when dividing 69354 by 9

- The Remainder when dividing 6×10000 , 9×1000 , 3×100 , 5×10 and 4 by 9 is the same as the remainder when dividing

$$6 \times 1 + 9 \times 1 + 3 \times 1 + 5 \times 1 + 4$$

This simplifies to $6 + 9 + 3 + 5 + 4 =$

Since 27 is divisible by 9, the number 69354 is also divisible by 9.

```
static boolean check(String s) {
    // Compute sum of digits
    int sum = 0;
    for (int i = 0; i < s.length(); i++)
        sum += (s.charAt(i) - '0');

    // Check if sum of digits is divisible by 9.
    return (sum % 9 == 0);
}

public static void main(String[] args) {
    String s = "69354";
    System.out.println(check(s) ? "Yes" : "No");
}
```

Checking if large Number Divisible by 11 or Not

Even - odd Digit Sum for Large String Input

Since input Number may be very large, we cannot use $n \% 11$ to check if a number is divisible by 11 or not. The idea is based on following fact

A Number is divisible by 11 if difference of following two is divisible by 11.

1. Sum of digits at odd places.
2. Sum of digits at even places.

For example, let us consider 76945

Sum of digits at odd places : $7 + 9 + 5$, Sum of digits at even places : $6 + 4$

Difference of two sums = $21 - 10 = 11$. Since difference is divisible by 11, the number 76945 is divisible by 11.

How does this work?

Let us consider 7694, we can write it as: $7694 = 7 \times 1000 + 6 \times 100 + 9 \times 10 + 4$

The proof is based on below observation:

- Remainder of 10^i divided by 11 = 1 if i is **even**
- Remainder of 10^i divided by 11 = 10 = -1 if i is **odd**

So the powers of 10 only result in values either 1 or -1. Remainder of " $7 \times 1000 + 6 \times 100 + 9 \times 10 + 4$ " divided by 11 can be written as :

$$7 \times (-1) + 6 \times 1 + 9 \times (-1) + 4 \times 1$$

The above expression is basically difference between sum of even digits and odd digits.

```
public static boolean divBy11(String s) {
    int n = s.length();
    int oddDigSum = 0, evenDigSum = 0;

    for (int i = 0; i < n; i++) {
        int digit = s.charAt(i) - '0';

        // check for even index
        if (i % 2 == 0)
            oddDigSum += digit;
        else
            evenDigSum += digit;
    }
    // Check divisibility by 11
    // using digit sum difference
    return (oddDigSum - evenDigSum) % 11 == 0;
}

public static void main(String[] args) {
    String s = "76945";
    System.out.println(divBy11(s) ? "true" : "false");
}
```